

Alex Levesque

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EDUCATION

- Queen’s University

BASc Applied Mathematics and Computer Engineering (GPA: 3.80/4.30)

• Coursework: Algorithms, Engineering Design, Probability, Real Analysis, Microeconomics, Philosophy
- Kingston, ON
Sep 2024 – Apr 2028
- University of Warwick

Incoming Exchange Semester

• Coursework: Data Science, Financial Markets, Time Series Analysis, Econometrics, Operations Research
- Coventry, UK
Jan 2027 – Apr 2027

AWARDS

- GRAMMY Music Award, The Recording Academy

• Received a **GRAMMY Award** for contributions to Killer Mike’s *MICHAEL*, 2024 Rap Album of the Year

• Produced for the Migos, Gunna, Peso Pluma, Roddy Ricch, Polo G, A Boogie, Lil Tjay, Skepta, Cam’ron
- Feb 2024
- Canada’s Top Students, Scotiabank

• Selected as 1 of 30 attendees (from 500+) for Scotiabank’s Canada’s Top Students conference and case competition
- Nov 2025

EXPERIENCE

- Data Science Intern

Scotiabank

• Designed and deployed an end-to-end LLM-based concurrent parsing pipeline for unstructured audit files, delivering **25%+ labor savings** and near-constant time complexity at scale using Vertex AI and Openpyxl

• Developed an interactive Power BI expense analysis dashboard surfacing **4 KPIs** across CAD spend and policy violations with dynamic 3-, 6-, and 12-month filtering, saving the expense team **~5 hours/week in manual review**

• Reduced **\$70M+ tech risk events loss exposure** by building a clustering pipeline with HDBSCAN, tuning parameters to reduce overlap, and improving efficiency with 66.7% fewer LLM calls
- May 2026 – Aug 2026
Toronto, ON
- Generative AI Engineering Intern

Government of Canada

• Designed and implemented Azure-based GenAI workflows to transcribe, categorize, and organize refugee documents achieving **93% accuracy**, reducing manual processing time, and improving operational efficiency

• Engineered an Azure chatbot leveraging an **ETL pipeline, Cosmos DB (NoSQL), sentiment analysis, and vector storage** to automate JSON legal document classification and presented results to stakeholders
- May 2025 – Aug 2025
Ottawa, ON

PROJECTS

For more projects and in-depth analyses regarding my projects, click here.

- Adaptive Reinforcement Learning Ensemble Strategy, Queen’s AI Club (QMIND)

• Implemented and trained an ensemble of **PPO, A2C, and TD3 reinforcement learning agents** for systematic portfolio allocation and position sizing across market regimes, with experiment tracking and version control via Git

• Designed a risk-aware diversification and rebalancing framework that reduced portfolio volatility and improved risk-adjusted returns, validated via Sharpe ratio, max drawdown, and turnover analysis against a SPY buy-and-hold
- Autonomous Multi-Agent Robotic Firefighting, Second Year School Project
- Developed and optimized Lloyd’s algorithm in MATLAB for adaptive **k-means clustering** of dynamic fire hotspot data, **reducing cluster convergence time by 30%** and improving autonomous robot deployment efficiency

• Leveraged GIS wildfire spatial data and analysis tools to delineate and model fire perimeters, enabling accurate real-time input for robotic firefighting cluster optimization and enhancing fire containment strategies

SKILLS & INTERESTS

Skills: Python, Excel, SQL, PowerBI, Microsoft Azure, Google Vertex AI, Cosmos DB, Openpyxl, Copilot Studio
ML/AI: PyTorch, Scikit-learn, HDBSCAN, CVXPY, Pandas, NumPy, Jupyter Notebook
Interests: Native in English and French, Karate Black Belt, Cycling, Medical Imaging, Psychology and Human Behavior